Customer Shopping Behavior Analysis

Retail & E-commerce Domain

Data Analytics Project (Python + SQL + Power Bi)

About me ➖ Soumyadeep Dhar
Passionate Data Analyst, transforming raw data into actionable insights.
Core Skills :- Advanced Excel, SQL, Power BI , Python, ETL
Currently doing Virtual Internship at AtliQ Technologies

AGENDA :- WHY? Objectives

WHAT ? Company's Details & Market

HOW? Data Request & Tools

Business Problem

A leading retail company is experiencing shifts in customer purchasing patterns across demographics, product categories, and sales channels, but lacks clarity on what is driving these changes. Without understanding the underlying factors such as discounts, seasonal trends, customer reviews, and payment preferences, the company is unable to effectively engage customers, retain loyalty, or optimize its marketing and product strategies. This gap in insight risks missed revenue opportunities and declining customer satisfaction.

Objectives

1. **Analyze Spending Patterns** — Examine transactional data from 3,900 purchases to understand how customers spend across product categories and demographics.
2. **Identify Purchase Drivers** — Determine which factors like discounts, reviews, seasons, and payment methods most significantly influence buying decisions and repeat purchases.
3. **Segment Customers** — Group customers by behavior, demographics, and preferences to enable personalized and targeted marketing.
4. **Evaluate Sales Channels** — Compare online vs. offline channel performance to optimize resource allocation and customer experience.
5. **Understand Subscription Behavior** — Analyze subscription patterns to improve retention strategies and long-term customer loyalty.
6. **Optimize Marketing and Product Strategy** — Translate data insights into actionable recommendations that improve customer engagement, drive sales, and support strategic business decisions.

You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Project Overview

This project analyzes customer shopping behavior using transactional data from **3,900 purchases** across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Deliverables

1. Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis.
2. Data Analysis (SQL): Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. Visualization & Insights (Power BI): Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
5. GitHub Repository: Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.

2. Dataset Summary

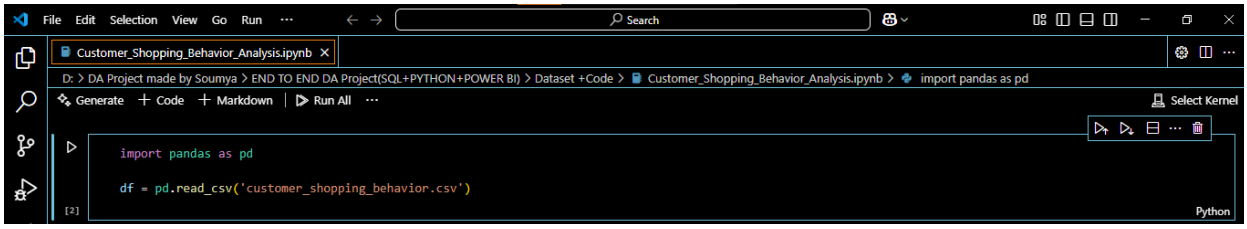
- Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating columns

3. **Exploratory Data Analysis using Python**

We began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using pandas.



● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

```python
df.describe(include='all')
```

[6] Python

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Pre Purc
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.0
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.3
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.4
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.0
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.0
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.0
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.0
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.0

Python

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

```
[7]  df.isnull().sum()
```

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

```python
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

[11]

```python
df.isnull().sum()
```

[12]

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

● **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

```python
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
```

[19] Python

```python
df.columns
```

[20] Python

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

● **Feature Engineering:**

○ Created age_group column by binning customer ages.

```python
# create a new column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['Age'], q=4, labels = labels)
```

[20] Python

```python
df[['Age','age_group']].head(10)
```

[21] Python

	Age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

○ Created purchase_frequency_days column from purchase data.

```python
# create new column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['Frequency of Purchases'].map(frequency_mapping)
```

[32] Python

● **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

```python
df[['Discount Applied','Promo Code Used']].head(10)
```

[37] Python

	Discount Applied	Promo Code Used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

● **Database Integration:** Connected Python script to MySQL and loaded the cleanedDataFrame into the database for SQL analysis.

```
# Code for MySQL
```
Python

```
[43] !pip install pymysql sqlalchemy
```
Python

```
Collecting pymysql
  Downloading pymysql-1.1.2-py3-none-any.whl.metadata (4.3 kB)
Requirement already satisfied: sqlalchemy in C:\Users\SOUMYADEEP\AppData\Local\Programs\Python\Python314\Lib\site-packages (2.0.46)
Requirement already satisfied: greenlet>=1 in C:\Users\SOUMYADEEP\AppData\Local\Programs\Python\Python314\Lib\site-packages (from sqlalchemy) (3.3.1)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\SOUMYADEEP\AppData\Local\Programs\Python\Python314\Lib\site-packages (from sqlalchemy) (4.15.0
Downloading pymysql-1.1.2-py3-none-any.whl (45 kB)
Installing collected packages: pymysql
Successfully installed pymysql-1.1.2

[notice] A new release of pip is available: 26.0 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

--Q1. What is the total revenue generated by male vs. female customers?

```
# Below you will find a comprehensive list of business questions and Queries

--Q1. What is the total revenue generated by male vs. female customers?

SELECT
    Gender,
    SUM(`Purchase Amount (USD)`) AS Total_Revenue
FROM customer
GROUP BY Gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Gender	Total_Revenue
Male	157890
Female	75191

--Q2. Which customers used a discount but still spent more than the average purchase amount?

```sql
--Q2. Which customers used a discount but still spent more than the average purchase amount?

SELECT `Customer ID`, `Purchase Amount (USD)`
FROM customer
WHERE `Discount Applied` = 'Yes'
  AND `Purchase Amount (USD)` >= (SELECT AVG(`Purchase Amount (USD)`) FROM customer);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer ID	Purchase Amount (USD)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76

-- Q3. Which are the top 5 products with the highest average review rating?

```sql
-- Q3. Which are the top 5 products with the highest average review rating?

SELECT `Item Purchased`, ROUND(AVG(`Review Rating`), 2) AS avg_rating
FROM customer
GROUP BY `Item Purchased`
ORDER BY avg_rating DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Item Purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

```sql
--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

SELECT `Shipping Type`, ROUND(AVG (`Purchase Amount (USD)`),2) AS avg_purchase
FROM customer
WHERE `Shipping Type` IN ('Standard', 'Express')
GROUP BY `Shipping Type`;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Shipping Type	avg_purchase
Express	60.48
Standard	58.46

--Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```sql
--Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

SELECT
    `Subscription Status`,
    COUNT(`Customer ID`) AS total_customers,
    ROUND(AVG (`Purchase Amount (USD)`),2) AS avg_spend,
    ROUND(SUM(`Purchase Amount (USD)`),2) AS total_revenue
FROM customer
GROUP BY `Subscription Status`
ORDER BY total_revenue,avg_spend DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Subscription Status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

--Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```sql
--Q6. Which 5 products have the highest percentage of purchases with discounts applied?

SELECT
    `Item Purchased`,
    COUNT(*) AS total_purchases,
    SUM(CASE WHEN `Discount Applied` = 'Yes' THEN 1 ELSE 0 END) AS discounted_purchases,
    ROUND(SUM(CASE WHEN `Discount Applied` = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS discount_pct
FROM customer
GROUP BY `Item Purchased`
ORDER BY discount_pct DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Toggle wrapping of cell contents

Item Purchased	total_purchases	discounted_purchases	discount_pct
Hat	154	77	50.00
Sneakers	145	72	49.66
Coat	161	79	49.07
Sweater	164	79	48.17
Pants	171	81	47.37

--Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```sql
--Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

WITH customer_type AS (
SELECT `Customer ID`, `Previous Purchases`,
    CASE
        WHEN `Previous Purchases` = 1 THEN 'New'
        WHEN `Previous Purchases` BETWEEN 2 AND 10 THEN 'Returning'
        ELSE 'Loyal'
    END AS customer_segment
FROM customer
)
SELECT customer_segment, count(*) as "Number pf Customers"
FROM customer_type
GROUP BY customer_segment;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_segment	Number pf Customers
Loyal	3116
Returning	701
New	83

--Q8. What are the top 3 most purchased products within each category?

```sql
--Q8. What are the top 3 most purchased products within each category?

WITH item_counts AS (
SELECT Category,
 `Item Purchased`,
 COUNT(`Customer ID`) AS total_orders,
  ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(`Customer ID`) DESC) AS item_rank
    FROM customer
    GROUP BY Category, `Item Purchased`
)
SELECT item_rank,category, `Item Purchased`, total_orders
FROM item_counts
WHERE item_rank <=3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

item_rank	category	Item Purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```sql
--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

SELECT
    CASE WHEN `Previous Purchases` > 5 THEN 'Repeat Buyer' ELSE 'Non-Repeat Buyer' END AS buyer_type,
    `Subscription Status`,
    COUNT(*) AS count
FROM customer
GROUP BY buyer_type, `Subscription Status`
ORDER BY buyer_type, `Subscription Status`;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

buyer_type	Subscription Status	count
Non-Repeat Buyer	No	329
Non-Repeat Buyer	Yes	95
Repeat Buyer	No	2518
Repeat Buyer	Yes	958

--Q10. What is the revenue contribution of each age group?

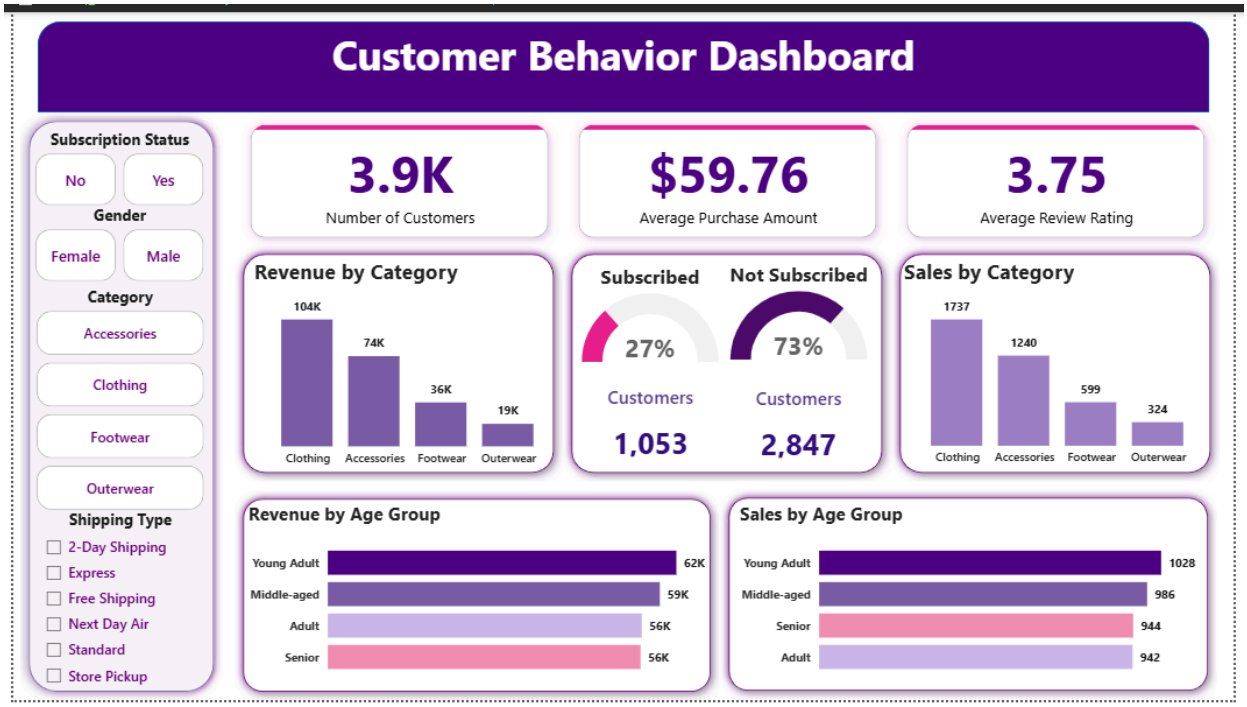
```sql
--Q10. What is the revenue contribution of each age group?

SELECT
    CASE
        WHEN Age < 18 THEN 'Under 18'
        WHEN Age BETWEEN 18 AND 25 THEN '18-25'
        WHEN Age BETWEEN 26 AND 35 THEN '26-35'
        WHEN Age BETWEEN 36 AND 50 THEN '36-50'
        WHEN Age BETWEEN 51 AND 65 THEN '51-65'
        ELSE '65+'
    END AS age_group,
    SUM(`Purchase Amount (USD)`) AS total_revenue,
    ROUND(SUM(`Purchase Amount (USD)`) * 100.0 / (SELECT SUM(`Purchase Amount (USD)`) FROM customer), 2) AS revenue_pct
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_group	total_revenue	revenue_pct
18-25	62143	26.66
51-65	59197	25.40
36-50	55978	24.02
51-65	55763	23.92

5. Dashboard in Power BI :-

Finally, we built an interactive dashboard in Power BI to present insights visually.



Business Recommendations :-

- Boost Subscriptions – Promote exclusive benefits for subscribers.

- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.

- Review Discount Policy – Balance sales boosts with margin control.

- Product Positioning – Highlight top-rated and best-selling products in campaigns.

- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.